The Limitations of SMOTE and Class Imbalance in Low-Sample Radiomics: A Cautionary Case Study in Renal Mass Differentiation

Abstract — **Purpose : Three-dimensional radiomics has been widely proposed as a non-invasive tool to differentiate malignant Renal Cell Carcinoma (RCC) from benign Angiomyolipoma (AML), a distinction that is often ambiguous on standard CT imaging due to the visual overlap of fat-poor AML variants with RCC. However, the reliability of radiomics pipelines built on small, severely imbalanced cohorts—a common scenario in rare or underrepresented pathologies—remains insufficiently scrutinized. This study investigates whether a 3D volumetric radiomics pipeline combining Support Vector Machine (SVM) classification with Synthetic Minority Over-sampling Technique (SMOTE) resampling can reliably differentiate RCC from AML in a small, highly imbalanced cohort (N = 69; 58 RCC, 11 AML), and more critically, whether single-fold or averaged performance metrics—as commonly reported in the radiomics literature—faithfully represent the stability of such a model.**

Methods — Using arterial-phase CT data from the UCSF RMaC dataset, 1182 volumetric radiomics features were extracted via PyRadiomics following isotropic resampling and intensity normalization. A nested cross-validation framework was implemented, with an inner loop performing SMOTE oversampling, ANOVA-based feature selection (SelectKBest), and grid-search hyperparameter tuning for an SVM classifier, and an outer loop providing five independent, leakage-free performance estimates.

Results — While the best-performing outer fold achieved a PR-AUC of 0.81, an accuracy of 94.1%, and an F1-score of 0.75, the full nested cross-validation revealed substantial volatility across folds, with AUC ranging from 0.24 to 0.81 and F1-score ranging from 0.0 to 0.75. This near four-fold swing in discriminative performance, depending solely on how 69 cases—and particularly the 11 minority-class cases—were partitioned, indicates that the model failed to learn a stable decision boundary.

Conclusion — Our findings demonstrate that standard oversampling techniques such as SMOTE fail to establish reliable decision boundaries when applied to small, high-dimensional medical imaging datasets, and that reporting a single "best" fold or an unweighted multi-fold average without transparently disclosing this volatility risks substantial performance inflation. We argue that feature-space interpolation among as few as 11 original minority vectors is structurally unable to capture the true variability of the minority class, and we propose that future radiomics augmentation strategies should shift from synthetic feature-vector generation toward image-level augmentation—including geometric transformations, intensity perturbation, and ROI boundary perturbation—which preserves genuine anatomical and spatial structure rather than interpolating in an abstract, decontextualized feature space.

Keywords — ***radiomics, renal cell carcinoma, angiomyolipoma, radiomics, computed tomography (CT), class imbalance,SMOTE, nested cross-validation, machine learning, support vector machines, decision boundry, data augmentation***

I. INTRODUCTION

In contemporary renal oncology, the accurate differentiation of renal masses is the cornerstone for determining appropriate medical or surgical pathways. The primary classification boundary lies between malignant Renal Cell Carcinoma (RCC) and benign Angiomyolipoma (AML). RCC is inherently invasive and metastatic, demanding aggressive interventions such as partial or radical nephrectomy, targeted therapies, or immunotherapies. Conversely, AML is a benign mesenchymal tumor that typically follows a conservative "watch and wait" surveillance protocol, rendering surgical intervention clinically unnecessary [1].

A severe clinical dilemma arises from the visual overlap between these two entities on standard computed tomography (CT) scans. Specifically, fat-poor AMLs (fp-AML) lack the macroscopic fat signatures typically used by radiologists for definitive diagnosis, causing them to look identical to RCCs during standard visual evaluations. Because human vision alone cannot reliably separate these sub-visual texture distributions, a high rate of unnecessary nephrectomies is performed on patients with completely benign disease [2].

Radiomics has been proposed as a means of moving "beyond the Hounsfield Unit," converting standard medical images into minable, high-dimensional feature spaces capable of revealing sub-visual biomarkers correlated with underlying tissue biology [3–7]. However, the rare and binary nature of pathologies such as AML means that radiomics cohorts are frequently small and severely class-imbalanced—precisely the conditions under which both feature extraction and resampling techniques are least theoretically justified. The vast majority of published radiomics studies report performance using k-fold cross-validation or, in many cases, a single train-test split, often without disclosing the variance across folds [13]. When a dataset is small and one class is rare, this practice creates a substantial risk: a single "lucky" partition of the data can produce an impressively high headline metric that does not represent the model's true, generalizable behavior.

This study uses the clinically important RCC-versus-AML differentiation task as a controlled case study to examine this broader methodological problem. Rather than presenting a single performance figure, we deliberately foreground the full distribution of outcomes obtained under rigorous nested cross-validation, and we use this distribution to investigate why a standard, widely adopted oversampling technique—SMOTE—fails to produce a stable decision boundary when only a handful of original minority-class examples are available. Our goal is not to dismiss radiomics as a diagnostic tool, but to demonstrate, transparently and quantitatively, the conditions under which its reported performance should be interpreted with caution, and to propose a more principled path—image-level data augmentation—toward genuinely stable models in low-sample, high-dimensional medical imaging tasks.

II. Materials and Methods

A. Collection Data and Cohort Selection

This study utilizes high-fidelity imaging data from the UCSF RMaC (University of California San Francisco 3D Multi-Phase Renal Mass CT Dataset with Tumor Segmentations) open-source repository [8]. To maintain the integrity of the original Hounsfield Units (HU) and avoid artifacts from manual manipulation, the data was utilized in its current high-fidelity state.

Cohort selection was handled via a deterministic filtering pipeline based on the dataset's provided metadata:

- Phase Selection: The entire cohort was filtered to isolate contrast-enhanced arterial phase images.
- Pathology Filtering: Images were stratified by confirmed histopathology to isolate Carcinoma and Angiomyolipoma.
- Final Sub-Cohort: This metadata-driven filtering yielded a final operational sub-cohort of 69 cases, comprising 11 angiomyolipoma cases and 58 carcinoma cases.

B. Radiomics Pipeline and Mathematical Foundations

The workflow extracts standardized 3D volumetric radiomics features across the entire segmented tumor mass, avoiding the loss of spatial data inherent to 2D single-slice boundaries. A total of 1182 features were extracted using the Python-based PyRadiomics toolkit. The computational implementation is built on a structured mathematical backbone where advanced matrices analyze spatial voxel interactions[9]:

- Image Standardization and Device Normalization:

Raw slice thicknesses in the UCSF RMaC cohort ranged from 0.625 mm to 10 mm, and in-plane spatial resolutions varied between 0.42 mm and 1.36 mm [2]. To overcome variability across different acquisition devices and ensure texture comparability, all volumes were resampled to a 1 mm × 1 mm × 1 mm isotropic spatial resolution using the SimpleITK image analysis toolkit in Python (version 3.12). Furthermore, to compensate for variations in Hounsfield Unit (HU) distributions across different scanners, intensity values were scaled using Z-score normalization [10].

- Mask Geometry and Phase Alignment:

To keep some of the images usable during the contrast-enhanced arterial phase, a registration tier (a multi-stage spatial alignment process) was implemented to resolve mismatches between image and mask geometry. For cases where dimensions varied, the arterial phase served as the fixed template, and the tumor masks were resized and coregistered using spline interpolation to preserve regional information and ensure the 3D volume accurately encapsulated the segmented tumor mass.

- Feature Categorization and Algorithmic Discovery:

The extracted feature space encompasses baseline 3D shape descriptors capturing macroscopic growth behaviors—such as sphericity, flatness, and elongation to differentiate the well-defined boundaries of benign angiomyolipomas (AML) from the chaotic, asymmetric margins of malignant renal cell carcinomas (RCC)—alongside advanced intensity and microstructural metrics. These metrics were computed both on the baseline images (including interpolation diagnostics) and across mathematically filtered spaces using Laplacian of Gaussian (LoG-sigma) filters at varying spatial scales and multi-directional wavelet decompositions. Within these spaces, first-order statistics quantified global gray-level distributions (e.g., mean, variance, skewness, and entropy) to evaluate baseline tumor density, while advanced second- and higher-order textural matrices—including the Gray-Level Co-occurrence Matrix (GLCM) and the Gray-Level Run Length Matrix (GLRLM)—calculated complex voxel-pair joint probabilities and consecutive intensity runs to mathematically chart structural tissue heterogeneity [9,11].

C. Downstream Machine Learning Designs

To address the severe class imbalance (~84% carcinoma vs. ~16% angiomyolipoma) resulting from the limited number of angiomyolipoma cases, a rigorous nested cross-validation framework is implemented . Because this limited dataset is not intended for immediate clinical deployment, the primary objective of this nested framework is to reliably evaluate and select the optimal machine learning algorithm to align with data profile, best features to overcome overfitting issues and balance computational resources for differentiating between these two distinct tumor types.

Within each outer fold of the nested cross-validation framework, test and training data are strictly isolated to prevent data leakage. In every inner loop, a sequential machine learning pipeline is executed independently on the training split, consisting of feature normalization via standard scaling, over-sampling via the Synthetic Minority Over-sampling Technique (SMOTE), feature selection via ANOVA-based ranking (analysis of variance to measure the statistical significance of feature differences across classes), and classification using a Support Vector Machine (SVM). Hyperparameter tuning is then performed via Grid Search analysis within each inner fold to simultaneously optimize the number of SMOTE nearest neighbors, the number of top features retained, the SVM kernel type (linear versus radial basis function), the regularization strength (C), and the kernel coefficient (gamma).

Using this pipeline, a variety of machine learning classifiers—including Random Forest, XGBoost, K-Nearest Neighbors (KNN), Logistic Regression, and Support Vector Machines (SVM)—were evaluated. The SVM consistently outperformed alternative classifiers on a per-fold basis and was retained as the focus of this analysis, both because of its theoretical suitability for boundary-focused classification in overlapping distributions [12,13] and because its outer-fold behavior most clearly exposes the instability investigated in this study.

Crucially, this workflow is identical in structure to what is reported as a success in much of the published radiomics literature: a multi-fold (or single-split) validation is run, a representative metric is computed, and that metric is reported as evidence of model validity. The purpose of the following sections is to show what this standard workflow conceals when it is not accompanied by full disclosure of per-fold variability.

III. Results

The proposed radiomics-based machine learning framework was evaluated using a nested cross-validation [14] strategy to ensure robustness under limited and imbalanced data conditions. Across the five outer folds, the Support Vector Machine (SVM) classifier's performance varied dramatically depending on how the 69-case cohort—and particularly the 11 angiomyolipoma cases—was partitioned between training and test sets.

As shown in Table I, the best-performing outer fold (Fold 2) achieved an AUC of 0.81, an accuracy of 94.1%, and an F1-score of 0.75, with optimal hyperparameters of SMOTE k_neighbors = 4, SelectKBest k = 7, and an SVC configured with kernel = 'rbf', C = 50, and gamma = 0.01. Considered in isolation, this fold would represent a strong, clinically promising result. However, the remaining four outer folds tell a markedly different story: Fold 4 achieved an AUC of only 0.24 and an F1-score of 0.0, correctly identifying zero of the five angiomyolipoma cases in its test partition, while Folds 1, 3, and 5 produced intermediate AUCs of 0.45, 0.56, and 0.39, respectively. This yields an outer-fold AUC range of 0.24–0.81—a swing of 0.57 AUC points—and an F1-score range of 0.0–0.75, depending entirely on which cases happened to fall into the training versus test partition for that fold. This wide dispersion highlights the extreme volatility of the decision boundary when subjected to different training/test splits of the same 69-case cohort.

Table I additionally reports the optimal hyperparameter configuration independently selected within each outer fold's inner loop. Notably, the selected SMOTE k_neighbors, SelectKBest k, and SVC kernel/C/gamma values are themselves unstable across folds—ranging from a linear kernel with k_neighbors = 6 in Fold 5 to an RBF kernel with k_neighbors = 4 in Fold 2—indicating that the inner-loop optimization is converging on substantially different solutions depending on which 11 minority-class cases (or fewer, within the training partition) are available to it.

IV. Discussion

The results of this study demonstrate that a standard, widely used radiomics workflow—SVM classification combined with SMOTE oversampling under nested cross-validation—fails to establish a stable decision boundary when applied to a small, severely imbalanced cohort. While an individual validation fold reached an AUC of 0.81, comparable to figures frequently reported as evidence of clinical promise in the radiomics literature, the full nested cross-validation distribution (range: 0.24–0.81) reveals that this figure reflects a favorable partition of the data rather than a generalizable model property.

A. *Why SMOTE Fails Under Severe Data Scarcity*

The mechanism underlying this volatility is rooted in how SMOTE constructs synthetic samples. SMOTE generates new minority-class instances by linearly interpolating between an original minority sample and one of its nearest neighbors within the same class [15]. In this study, that interpolation pool was restricted to the 11 original angiomyolipoma cases—reduced further to as few as 8–9 cases within any given training fold once a test partition was removed. In a feature space of 1182 dimensions, 8–11 points are extraordinarily sparse: they cannot meaningfully sample the true underlying distribution of angiomyolipoma imaging phenotypes, and the synthetic samples generated by interpolating between them are constrained to lie along a small number of line segments embedded within that high-dimensional space.

This produces what can be described as synthetic noise clusters: tight, artificial groupings of interpolated points that reflect the idiosyncrasies of whichever 8–11 original cases happened to fall into the training partition, rather than the true biological variability of fat-poor AML. Because the composition of this small pool shifts with every cross-validation split, the synthetic minority-class region of the feature space shifts correspondingly—and with it, the position of the SVM decision boundary. This is consistent with theoretical and empirical work showing that SMOTE's benefits, well established in low-dimensional or moderately sized datasets, do not reliably extend to high-dimensional, small-sample regimes, where the technique can fail to attenuate majority-class bias and may even introduce spurious correlation structure into the augmented data without changing the class-specific mean values, while decreasing data variability [21]. Related work using independent radiomics cohorts has likewise shown that single train-test splits, and even repeated cross-validation, can produce highly unstable area-under-curve estimates when sample sizes are small and the classification task is difficult, with training-test AUC gaps and absolute performance varying substantially across resampled partitions of the same dataset [22]. Our findings extend this body of evidence to the specific case of feature-space oversampling under severe class imbalance, and reinforce the conclusion that the SVM, despite its theoretical strength as a margin-maximizing, boundary-focused classifier [12,13], is only as stable as the data used to define its support vectors—and 11 original cases are simply not enough to anchor that boundary reliably.

B. *The risk of Performance Inflation in the Radiomics Literature*

A broader implication of these findings concerns reporting practices across the radiomics field. Many published studies report a single train-test split or a multi-fold average without disclosing the per-fold or per-split variance, and some effectively select or emphasize the most favorable validation result [13]. Our Fold 2 result (AUC = 0.81, accuracy = 94.1%, F1 = 0.75) is, in isolation, indistinguishable from the kind of headline metric that populates much of the published literature on small-cohort radiomics classification. Only by exposing the full nested cross-validation distribution—and explicitly contrasting the "lucky" and "unlucky" folds—does the instability of the underlying model become visible. We argue that this transparent reporting practice, rather than the pursuit of a single optimized metric, should be the standard expected of radiomics studies operating under similarly constrained data conditions.

C. Toward Image-Level Data Augmentation

These findings motivate a shift in how data scarcity is addressed in small radiomics cohorts. SMOTE, and tabular oversampling techniques more broadly, operate by interpolating blindly within an abstract, decontextualized mathematical feature space—a space that no longer retains any direct correspondence to anatomical structure once the original CT volume has been reduced to a vector of 1182 numbers. By contrast, augmentation performed directly at the image level preserves actual spatial anatomy and pixel-level relationships, generating new training examples that remain physically and biologically plausible rather than purely numerically interpolated.

Image-level manipulations relevant to this pipeline fall into two distinct categories:

- Geometric and Intensity Manipulations: standard computer vision augmentations such as 3D random rotations, translations, and scaling of the CT volume, together with the addition of controlled Gaussian noise to the Hounsfield Unit distribution to simulate variability across different CT scanner models and acquisition protocols.
- ROI (Region of Interest) Perturbations: small, controlled modifications to the tumor segmentation boundary itself—such as eroding the contour by one voxel, dilating it, or applying contour randomization—which force the radiomics pipeline to extract features from slightly different, clinically plausible variations of the same underlying tumor, thereby testing and building feature and model stability rather than introducing synthetic statistical noise.

ROI perturbation strategies of this kind have already been validated in adjacent radiomics applications as a means of assessing and improving feature robustness under realistic segmentation variability [16], and similar morphological perturbation chains—combining contour randomization with controlled volume growth and shrinkage—have been shown to reliably distinguish robust from non-robust radiomic features without requiring repeat imaging [20]. Unlike SMOTE, which generates entirely synthetic feature vectors with no image-domain referent, these perturbations generate new feature vectors that correspond to genuinely plausible renderings of the same physical tumor, anchoring any resulting augmentation in real anatomical variability rather than statistical extrapolation from a handful of cases.

D. Limitation

Several limitations qualify these findings. First, the cohort size (N = 69, 11 AML cases) is itself a primary subject of this investigation rather than an incidental constraint, and the specific volatility magnitudes reported here are not necessarily representative of all small-cohort radiomics tasks; the underlying mechanism, however—interpolation among a sparse minority pool in high-dimensional space—generalizes to any comparably scarce cohort. Second, this study relies on a single institutional dataset and a single imaging phase (arterial); external validation across additional scanners, protocols, and institutions remains necessary before any conclusions about generalizable RCC/AML differentiation performance can be drawn, independent of the stability question addressed here. Third, while the proposed shift toward image-level augmentation is theoretically well-motivated and supported by prior robustness literature, it has not yet been empirically implemented or validated within this specific cohort; its capacity to stabilize the SVM decision boundary in this RCC/AML task remains a hypothesis for future work rather than a demonstrated result.

V. Conclusion and Future Work

This study set out to differentiate Renal Cell Carcinoma from Angiomyolipoma using a 3D volumetric radiomics pipeline under severe class imbalance, and in doing so, surfaced a more consequential methodological finding: standard tabular oversampling via SMOTE does not produce a stable decision boundary when the minority class is represented by as few as 11 original cases embedded in a 1182-dimensional feature space. While individual validation folds reached an AUC as high as 0.81, rigorous nested cross-validation exposed a performance range of 0.24–0.81 AUC across outer folds—a volatility that would remain invisible under conventional single-split or averaged reporting practices common in the radiomics literature.

Given the volatility observed when applying tabular oversampling (SMOTE) to high-dimensional feature spaces with severe class scarcity, future work will pivot toward image-level data augmentation. Rather than generating synthetic mathematical vectors, we aim to implement advanced image manipulations—including 3D geometric transformations, Hounsfield Unit intensity shifting, and ROI boundary perturbations (erosion and dilation)—directly on the source CT volumes. Generating synthetic data at the image acquisition level rather than the feature extraction level is expected to preserve localized spatial texture and structural anatomy more effectively, ultimately stabilizing the SVM decision boundary and improving cross-validation generalizability.

Beyond image-level augmentation, future expansions of this work will involve accessing new, diverse data sources and implementing multi-phase analysis to assess whether the instability observed here is specific to the arterial phase or persists across imaging phases. Future iterations will also benchmark alternative tabular and algorithmic resampling strategies—including Borderline-SMOTE, ADASYN, and Balanced Random Over-Sampling (ROS) with Noise, alongside generative architectures such as Generative Adversarial Networks (GANs) and Diffusion Models [18,19]—directly against the image-level augmentation approach proposed here, to determine which strategy most effectively stabilizes classification performance under severe data scarcity. Finally, the publication of this framework and its cautionary findings is intended to catalyze multidisciplinary collaboration within the local and universal urology communities [17], encouraging more transparent reporting of cross-validation variance and facilitating access to the larger, more diverse datasets ultimately required to validate any radiomics-based diagnostic tool for clinical use.

TABLE I : NESTED CROSS-VALIDATION PERFORMANCE AND OPTIMAL HYPERPARAMETER CONFIGURATIONS

Outer Fold	Metrics	Smote	SelectKBest	SVC	SVC	SVC	Confussion Matrix	
	AUC ACCURACY F1-SCORE	k_neighbors:	k:	C:	gamma:	kernel:	TN	FP
							FN	TP
1	AUC : 0.45 ACC. : 91.1% F1-SCORE : 0.66	6	17	0.5	‘auto’	‘rbf’	28	1
							2	3
2	AUC : 0.81 ACC. : 94.1% F1-SCORE: 0.75	4	7	50	0.01	‘rbf’	29	0
							2	3
3	AUC : 0.56 ACC.: 88.2% F1-SCORE: 0.50	4	9	5	0.1	‘rbf’	28	1
							3	2
4	AUC : 0.24 ACC. : 75.7% F1-SCORE : 0.0	5	7	5	0.1	‘rbf’	25	3
							5	0
5	AUC : 0.39 ACC: 84.8% F1-SCORE: 0.54	6	17	5	‘scale’	‘linear’	25	3
							2	3

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REFERENCES

[1] I. A. Kawalit, Z. H. Kalaji, A. M. Ibrahim, E. M. Emtair, and L. F. Al-Rabati, “Multidisciplinary Approach to a Complicate Renal Angiomyolipoma: A Case Report Review”, Medical Research Archives, vol. 12, no. 10, October 2024, doi://doi.org//10.18103/mra.v12i10.5880.

[2] F.Dehghani Firouzabadi, et al., “CT radiomics for differentiating fat poor angiomyolipoma from clear cell renal cell carcinoma: Systematic review and meta-analysis,” PLOS ONE, doi: 10.1371/journal.pone.0287299, 2023.

[3] S.H. Moon, J. Kim, J-G Joung, et al, “Correlations between metabolic texture features, genetic heterogeneity, and mutation burden in patients with lung cancer”, Eur J Nucl Med Mol Imaging, 2019, 46, 446-454, doi: 10.1007/s00259-018-4138-5

[4] N. Altinok and A. Guvenis, “Interpretable radiomics method for predicting human papillomavirus status in oropharyngeal cancer using Bayesian networks,” Physica Medica: European Journal of Medical Physics, vol. 114, doi: 10.1016/j.ejmp.2023.102671, 2023.

[5] K. Sarac and A. Guvenis, “Using radiomics for predicting the HPV status of oropharyngeal tumors,” Journal of Engineering and Applied Science, vol. 71, no. 11, doi: 10.1186/s44147-023-00355-w, 2024.

[6] H. Beste Okmen, A. Guvenis, and H. Uysal, “Predicting the polybromo-1 (PBRM1) mutation of a clear cell renal cell carcinoma using computed tomography images and KNN classification with random subspace,” Vibroengineering Procedia, vol. 26, doi: 10.21595/vp.2019.20931, 2019.

[7] L. Russo, D. Charles-Davies, S. Bottazzi, L. Boldrini, “Radiomics for clinical decision support in radiation oncology”, Clinical Oncology, vol.36,pp. 269-281, 2024, doi: 10.1016/j.clon.2024.03.003

[8] S. Sahin , E. Diaz, A. Rajagopal, et al., “UCSF RMaC: University of California San Francisco 3D Multi-Phase Renal Mass CT Dataset with Tumor Segmentations,” medRxiv preprint, doi: 10.64898/2026.02.11.26346096, 2026.

[9] M.E. Mayerhoefer, et al., “Introduction to Radiomics”, The Journal of Nuclear Medicine, vol. 61, April 2020, DOI: 10.2967/jnumed. 118.222893.

[10] H. Karagoz and A. Guvenis, “Predicting the Grade of Clear Cell Renal Cell Carcinoma from CT Images Using Random Subspace-KNN and Random Forest Classifiers,” in Proceedings of USBEREIT, doi: 10.1109/USBEREIT51232.2021.9455032, 2021.

[11] S. Rizzo, et al., “ Radiomics: the facts and the challenges of image analysis”, European Radiology Experimental 2, 36 (2018), doi : 10.1186/s41747-018-0068-z.

[12] Y. Lee, “Support Vector Machines for Classification: A Statistical Portrait,” in Statistical Methods in Molecular Biology (Methods in Molecular Biology, Vol. 620), H. Bang et al., Eds. New York, NY: Springer Science+Business Media, LLC, 2010, pp. 347–368. doi: 10.1007/978-1-60761-580-4_11.

[13] C. Cortes and V. Vapnik, “Support-vector network,, “Machine Learning, vol.20, no. 3, pp. 273-297, 1995, doi: 10.1007/BF00994018.

[14] S. Parvandeh, H-W. Yeh, M. P. Paulus, B. A. McKinney, “Consensus features nested cross-validation”, Bioinformatics, 36(10), 2020, 3093-3098, doi: 10.1093/bioinformatics/btaa046.

[15] H. Karagoz and A. Guvenis, “Predicting the Grade of Clear Cell Renal Cell Carcinoma from CT Images Using Random Subspace-KNN and Random Forest Classifiers,” in Proceedings of USBEREIT, doi: 10.1109/USBEREIT51232.2021.9455032, 2021.

[16] G. Lo Iacono, et al., “A Novel Data Augmentation Method for Radiomics Analysis Using Image Perturbations,” Journal of Imaging Informatics in Medicine, vol. 37, pp. 2401–2414, doi: 10.1007/s10278-024-01013-0, 2024.

[17] L.Han, et al., “Fully automated segmentation and classification of renal tumors on CT scans via machine learning,” BMC Cancer, vol. 25, no. 173, doi: 10.1186/s12885-025-13582-6, 2025.

[18] B. Kocak, E. S. Durmaz, E. Ates, &O. Kilickesmez, “Radiomics with artificial intelligence: a practical guide for beginners” Diagnostic and Interventional Radiology, 2019, doi: 10.5152/dir.2019.19321.

[19] A. G. Udu, M. T. Salman, M. K. Ghalati, A. Lecchini-Visintini, D. R. Siddle, H. Dong, “Emerging SMOTE and GAN Variants for Data Augmentation in Imbalance Machine Learning Tasks: A Review”, IEEE Access, vol. 13, pp 113838-11853, doi: 10.1109/ACCESS.2025.3584532

[20] F. Lo Iacono, G. Pontone, and V. D. A. Corino, “Assessing Left Ventricle Radiomic Features Robustness by Segmentation Perturbations,” in MEDICON’23 and CMBEBIH’23, IFMBE Proceedings, vol. 94, Springer, Cham, 2024, doi: 10.1007/978-3-031-49068-2_36.

[21] R. Blagus and L. Lusa, “SMOTE for high-dimensional class-imbalanced data,” BMC Bioinformatics, vol. 14, no. 106, 2013, doi: 10.1186/1471-2105-14-106.

[22] J. E. Park, et al., “Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results,” PLOS ONE, vol. 16, no. 8, e0256152, 2021, doi: 10.1371/journal.pone.0256152.

[23] K. Vrettos, M. Triantafyllou, et al., “Artificial intelligence-driven radiomics: developing valuable radiomics signature with use of artificial intelligence”, BJR|Artificial Intelligence, Volume 1, Issue 1, January 2024, doi: 10.1093/bjrai/ubae011.